

Strong-Field Regime of Double Spacetime Theory: Density-Dependent Cosine Potential and Layered Torsional Structures

<https://github.com/hypernumbernet>

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Abstract

The double spacetime theory reformulates gravity as torsional mismatch between particle-intrinsic usual and dual spacetimes, exactly reproducing General Relativity via teleparallel equivalence while eliminating the continuum hypothesis. In the strong-field regime, however, the theory requires a mechanism to drive the dual rotation angle ϕ to dominate over the usual rapidity θ , producing alternating attractive-repulsive layers. This paper proposes a microscopic density-dependent cosine potential governing the relative rotor phase, analogous to Josephson coupling in superconductivity. The resulting dynamics naturally generate scale-invariant layered structures, unifying nuclear forces with gravity, resolving singularities in torsion stars, and providing testable predictions for nuclear matter and compact objects.

1 Introduction

The double spacetime theory [?] provides a radical reformulation of gravitation, interpreting it not as curvature of a continuous manifold but as torsional mismatch between two particle-intrinsic spacetimes encoded in the 16-real-dimensional biquaternion algebra isomorphic to $\text{Cl}(3, 1)$. The usual spacetime, with basis (j, kI, kJ, kK) , governs standard kinematics, while the dual spacetime, with basis (k, jI, jJ, jK) , is related by right-multiplication by the pseudoscalar i , reversing the arrow of time while preserving the Minkowski norm.

Lorentz transformations and parallel transport are unified via commuting rotors $R_{\text{usual}} = \exp(\theta/2 \hat{p})$ ($\hat{p}^2 = +1$, boost-like) and $R_{\text{dual}} = \exp(\phi/2 \hat{q})$ ($\hat{q}^2 = -1$, rotation-like). The complete rotor is $R_{\text{total}} = R_{\text{usual}} R_{\text{dual}}$, and the torsional mismatch rotor $\Omega = R_{\text{usual}}^\dagger R_{\text{dual}}$ yields the bivector $\Omega_{\text{biv}} = \log \Omega$. The unique Lorentz-invariant scalar is the Killing form

$$J = \frac{1}{16} B(\Omega_{\text{biv}}, \Omega_{\text{biv}}).$$

The action $S = \frac{c^4}{16\pi G} \int J d^4x$ is dynamically equivalent to the Einstein-Hilbert action (up to boundary terms), reproducing General Relativity exactly as the biquaternionic realization of teleparallel gravity (TEGR) [?].

This framework eliminates Christoffel symbols, Riemann curvature, and the continuum hypothesis altogether, demoting spacetime to paired, compactified attributes of particles themselves. Inertia and gravity unify as the same dual-rotor rigidity phenomenon, dark matter becomes superfluous via logarithmic torsional memory, and singularities are forbidden in favor of layered torsion stars.

While the theory excels in the weak-field and cosmological regimes, the strong-field limit—nuclear densities, neutron stars, and beyond—poses a key open question: how does the dual rotation angle ϕ grow to dominate the usual rapidity θ as density increases? Weak fields are characterized by $\theta \gg \phi \approx \theta$ (slight mismatch yielding $J > 0$, attraction), but extreme densities require $\phi \gg \theta$ to flip the sign of J ($J < 0$, repulsion) and generate the predicted infinite alternating attractive-repulsive layers.

This paper addresses this challenge by proposing a microscopic, density-dependent cosine potential that governs the relative phase between usual and dual rotors. Inspired by Josephson coupling in superconductivity, the potential introduces collective resonance among particle-intrinsic dual spacetimes without altering the macroscopic definition of J or the action. The resulting dynamics naturally produce scale-invariant layered torsional structures, unifying the strong nuclear force with gravity as successive

cycles of the same mechanism, reinterpreting atomic nuclei as primordial torsion stars, and providing a self-regulating endpoint for collapse.

The structure of the paper is as follows. Section 2 briefly reviews the double spacetime framework and highlights the strong-field open problem. Section 3 introduces the cosine potential and derives its equations of motion. Section 4 details the density dependence of the coupling strength. Section 5 explores the deep analogy with superconductivity. Sections 6 and 7 discuss implications for nuclear physics, compact objects, and observational tests. We conclude with outlook on gravitational engineering and quantum extensions.

This extension preserves the algebraic elegance and background independence of the original theory while illuminating its rich strong-field phenomenology.

2 Review of Double Spacetime Theory

The double spacetime theory [?] reinterprets gravitation as a torsional phenomenon arising from the relative misalignment between two compactified Minkowski spacetimes intrinsically attached to every massive particle. The mathematical foundation is the 16-real-dimensional biquaternion algebra, generated by two commuting copies of Hamilton’s quaternions (i, j, k) and (I, J, K) , which is canonically isomorphic to the Clifford algebra $\text{Cl}(3, 1)$ with Minkowski signature $(-, +, +, +)$.

The usual spacetime is spanned by the basis (j, kI, kJ, kK) , representing vectors of the form

$$X = ct\,j + x\,kI + y\,kJ + z\,kK,$$

with invariant norm $X^2 = -(ct)^2 + x^2 + y^2 + z^2$. The dual spacetime, related by right-multiplication by the pseudoscalar i (reversing the time arrow), has basis (k, jI, jJ, jK) and vectors

$$X' = ct'\,k + x'\,jI + y'\,jJ + z'\,jK,$$

preserving the same norm $X'^2 = X^2$.

Lorentz boosts and rotations are generated by unit bivectors: $\hat{p} = p_1 iI + p_2 iJ + p_3 iK$ ($\hat{p}^2 = +1$, boost-like in the usual sector) and $\hat{q} = q_1 I + q_2 J + q_3 K$ ($\hat{q}^2 = -1$, rotation-like in the dual sector). The separate rotors are

$$R_{\text{usual}} = \exp\left(\frac{\theta}{2}\hat{p}\right) = \cosh\frac{\theta}{2} + \sinh\frac{\theta}{2}\hat{p},$$

$$R_{\text{dual}} = \exp\left(\frac{\phi}{2}\hat{q}\right) = \cos\frac{\phi}{2} + \sin\frac{\phi}{2}\hat{q},$$

yielding the complete rotor $R_{\text{total}} = R_{\text{usual}}R_{\text{dual}}$.

The torsional mismatch is captured by

$$\Omega = R_{\text{usual}}^\dagger R_{\text{dual}}, \quad \Omega_{\text{biv}} = \log \Omega,$$

and the unique quadratic Lorentz-invariant scalar is the Killing form

$$J = \frac{1}{16}B(\Omega_{\text{biv}}, \Omega_{\text{biv}}).$$

Positive J corresponds to usual-sector (boost) dominance and attractive gravity, while negative J arises from dual-sector (rotation) dominance and repulsion.

The action

$$S = \frac{c^4}{16\pi G} \int J\,d^4x$$

is identical (up to boundary terms) to the torsion scalar T of teleparallel equivalent of general relativity (TEGR) [?], yielding field equations exactly equivalent to Einstein’s. All classical predictions of General Relativity—Newtonian limit, post-Newtonian parameters, black hole solutions, cosmology, and gravitational waves—are reproduced precisely, without Christoffel symbols, Riemann tensor, or a priori continuum manifold.

The theory further unifies inertia with gravity as dual-rotor rigidity, eliminates dark matter via long-range torsional phase accumulation, and forbids singularities by predicting layered torsion stars with alternating attractive-repulsive zones.

Despite this success in weak and intermediate regimes, the strong-field limit remains incompletely specified within the purely algebraic framework. At nuclear and super-nuclear densities, the theory predicts infinite sign oscillations of J producing self-regulating layered structures, but the microscopic mechanism driving the dual angle ϕ to rapidly exceed the usual rapidity θ —flipping dominance from boost-like to rotation-like contributions in Ω_{biv} —is not prescribed by the rotors alone. The following sections address this gap by introducing a density-dependent cosine potential that provides the necessary collective dynamics while preserving the macroscopic equivalence with GR.

3 The Cosine Potential Hypothesis

To explain the transition from usual-spacetime dominance to dual-spacetime dominance in the strong-field regime, we introduce a microscopic effective potential that governs the relative alignment between the usual rapidity θ and the dual rotation angle ϕ at the level of individual particles. This potential is motivated by the need for a collective mechanism that amplifies dual-sector excitations at high densities without modifying the macroscopic definition of the torsional mismatch scalar J .

Consider a massive particle in a local environment characterized by baryon or energy density ρ . The proposed effective potential is

$$V(\theta, \phi; \rho) = \frac{m}{2}\theta^2 - \mu(\rho)[1 - \cos(\phi - \theta)],$$

where

- $m > 0$ is the rest mass, encoding the rigidity of the usual sector (the quadratic term drives attractive mismatch in the weak-field limit),
- $\mu(\rho) \geq 0$ is a density-dependent coupling strength that measures collective resonance among neighboring particles' dual spacetimes,
- The cosine term exploits the intrinsic compactness of the dual sector ($\phi \bmod 4\pi$), favoring discrete phase differences $\Delta\phi = \phi - \theta \approx \pi + 2\pi n$ ($n \in \mathbb{Z}$) when μ is large.

This form ensures compatibility with the vacuum limit: as $\rho \rightarrow 0$, $\mu \rightarrow 0$, and the minimum lies at $\phi \approx \theta$ (perfect anti-synchronization, $J \approx 0$). At finite but low density, small deviations $\Delta\phi \ll 1$ yield $1 - \cos(\Delta\phi) \approx (\Delta\phi)^2/2$, recovering weak attractive torsion proportional to mass density.

The equations of motion derived from this potential (supplementing the full rotor dynamics) are

$$\begin{aligned}\ddot{\theta} &= m\theta + \mu(\rho)\sin(\phi - \theta), \\ \ddot{\phi} &= -\mu(\rho)\sin(\phi - \theta).\end{aligned}$$

In static, spherically symmetric configurations, these reduce to first-order gradient equations coupled to the density profile $\rho(r)$.

At high density ($\mu \gg m\theta$), the deep cosine wells dominate, locking $\Delta\phi$ near odd multiples of π . This forces rapid growth of ϕ (multi-winding) to minimize the potential, shifting dominance in $\Omega_{\text{biv}} = \log(R_{\text{usual}}^\dagger R_{\text{dual}})$ from boost-like ($\hat{p}$, positive Killing contribution) to rotation-like ($\hat{q}$, negative Killing contribution). The resulting sign flip of J produces repulsive layers, followed by successive windings accessing deeper minima and generating alternating attractive zones—yielding the infinite layered structure characteristic of torsion stars.

Crucially, this hypothesis requires *no alteration* to the algebraic computation of J or the action $S = \frac{c^4}{16\pi G} \int J d^4x$. The cosine potential merely selects the density-dependent values of $\theta(r)$ and $\phi(r)$ that are fed into the unchanged rotor product $R_{\text{total}} = R_{\text{usual}} R_{\text{dual}}$. In the continuum weak-field limit, $\mu(\rho)$ becomes negligible, preserving exact equivalence with TEGR and General Relativity.

The following section specifies the functional form of $\mu(\rho)$ and its physical justification.

4 Density-Dependent Coupling $\mu(\rho)$

The strength of the cosine potential is controlled by the coupling parameter $\mu(\rho)$, which encodes the intensity of collective resonance among the compact dual spacetimes of neighboring particles. As particle separation decreases with increasing density ρ , the overlap or “tunneling” between individual dual sectors grows, amplifying the effective Josephson-like coupling between their rotation angles ϕ .

A physically well-motivated functional form is the power-law

$$\mu(\rho) = \mu_0 \left(\frac{\rho}{\rho_c} \right)^\gamma,$$

where

- $\rho_c \approx 2.8 \times 10^{14} \text{ g/cm}^3$ (or equivalently $0.17 \text{ nucleons/fm}^3$) is the nuclear saturation density, marking the onset of the first repulsive torsional layer (hard-core repulsion),
- μ_0 is a fundamental coupling scale with dimensions of energy (or mass in natural units), expected to be suppressed relative to the Planck scale but tuned to reproduce observed nuclear binding and neutron-star properties,
- $\gamma > 0$ determines the sharpness of the transition from weak to strong dual-sector coupling.

The power-law form is suggested by several considerations:

- Mean-field pair interactions yield $\gamma \approx 1\text{--}2$, as each particle resonates with a number of neighbors proportional to local density.
- Multi-body effects and Pauli exclusion in fermionic matter favor higher powers; $\gamma = 4$ mirrors the quartic term in Ginzburg–Landau theory and produces rapid onset of phase locking suitable for the sharp nuclear hard core.
- Scale invariance of the underlying torsional layers across density regimes naturally accommodates a pure power law without additional dimensionful scales.

We recommend $\gamma = 4$ as the baseline, yielding explosive growth of μ beyond ρ_c and facilitating multi-winding configurations ($n \gg 1$) at supra-nuclear densities. Lower values ($\gamma = 2$) produce smoother transitions, while $\gamma > 4$ sharpens the layer boundaries further.

Alternative forms may be considered for specific regimes:

- Exponential: $\mu(\rho) = \mu_0 \exp\left[\kappa \left(\frac{\rho}{\rho_c} - 1\right)\right]$ for $\rho > \rho_c$, reflecting tunnel-barrier penetration analogous to Josephson critical current in thin insulating layers.
- Fermi-like smoothing: $\mu(\rho) = \frac{\mu_0}{1 + \exp[-(\rho - \rho_c)/\Delta\rho]}$, introducing a finite transition width $\Delta\rho$ for numerical stability.

In the weak-field limit ($\rho \ll \rho_c$), $\mu(\rho) \approx 0$, rendering the cosine term negligible and recovering the standard linear torsional response of the original theory. At nuclear saturation, $\mu(\rho_c) = \mu_0$, calibrated such that the first minimum at $\Delta\phi \approx \pi$ reproduces the observed short-range repulsion. In neutron-star interiors ($\rho \sim 10^{15}\text{--}10^{18} \text{ g/cm}^3$), $\mu \gg \mu_0$ drives successive windings, generating the predicted infinite layered structure.

The precise value of μ_0 and optimal γ will be constrained by fitting nuclear equation-of-state data and neutron-star mass-radius relations in forthcoming numerical studies. The power-law form ensures background independence and compatibility with the particle-local nature of dual spacetimes, as $\mu(\rho)$ emerges statistically from ensemble averaging over discrete particle positions.

The deep structural similarity to Josephson coupling and superconducting order parameters is explored in the next section.

5 Analogy with Superconductivity and Josephson Junctions

The cosine potential introduced in Section 3 bears a striking structural resemblance to the Josephson energy across a weak link in superconductivity, suggesting a profound analogy between dual-spacetime resonance and macroscopic quantum coherence. This correspondence is not superficial: the compact topology of the dual sector ($\text{SU}(2) \cong \text{S}^3$) mirrors the phase degree of freedom of the superconducting order parameter, while density-dependent coupling $\mu(\rho)$ plays the role of tunnel-mediated pair coherence.

In a Josephson junction, two superconductors separated by a thin insulator are described by order parameters $\psi_1 = |\psi|e^{i\theta_1}$ and $\psi_2 = |\psi|e^{i\theta_2}$. The coupling energy is

$$E_J[1 - \cos(\theta_1 - \theta_2)],$$

where E_J grows with decreasing barrier thickness (increasing tunnel probability). This exactly parallels our potential

$$-\mu(\rho)[1 - \cos(\phi - \theta)],$$

with mappings:

- Usual rotor phase $\theta \leftrightarrow \theta_1$ (driven by external “voltage” analogous to acceleration or gravitational rapidity),
- Dual rotor phase $\phi \leftrightarrow \theta_2$ (compact and multi-winding-capable),
- $\mu(\rho) \leftrightarrow E_J$ (enhanced by denser particle packing, increasing dual-sector “tunneling”).

Key phenomenological analogies include:

- **Phase locking:** At strong coupling ($\mu \gg m\theta$), the relative phase $\Delta\phi = \phi - \theta$ locks to minima of the cosine well, preferentially at odd multiples of π for maximal depth. This mirrors DC Josephson current flow with zero phase difference in the ground state, but here the π -shift produces torsional repulsion ($J < 0$).
- **Plasma oscillations:** Applying a voltage bias in a junction excites phase oscillations at the Josephson plasma frequency $\omega_p \propto \sqrt{E_J}$. Similarly, perturbations in θ (e.g., infall or acceleration) drive $\Delta\phi$ oscillations at frequency $\sqrt{\mu(\rho)}$, manifesting as longitudinal torsional waves in dense matter—potential sources of pulsar radio emission and gravitational-wave echoes from layer transitions.
- **Flux quanta and winding numbers:** In type-II superconductors, magnetic flux penetrates as vortices carrying quantized circulation $\oint \nabla\phi dl = 2\pi n$. The dual sector’s compactness allows analogous multi-winding configurations ($n \in \mathbb{Z}$) as density increases, with each successive 2π winding corresponding to penetration into deeper torsional layers (alternating attraction/repulsion). These “torsional vortices” regulate collapse, forbidding singularities in the same way Abrikosov vortices distribute flux.

This analogy elevates high-density nuclear and supra-nuclear matter to the status of a *torsional superconductor*: a macroscopic coherent state where countless particle-intrinsic dual spacetimes lock into a collective phase, exhibiting “superfluid” resistance to torsional shear (repulsive barriers) and quantized excitations (layered structure). Atomic nuclei emerge as primordial torsional superconductors with the first winding cycle, while neutron-star interiors realize higher winding numbers in a chaotic, turbulent condensate.

The interpretive power of this analogy extends to experimental probes: phenomena such as critical densities (analogous to H_{c1} , H_{c2}), hysteresis in layer transitions, and coherent emission (torsional Shapiro steps under external fields) become predictable. It also motivates gravitational engineering via artificial enhancement of μ using superconducting metamaterials to mimic high-density resonance.

The physical consequences of this torsional superconductivity for nuclear forces and compact objects are explored in the following section.

6 Consequences for Nuclear Physics and Compact Objects

The density-dependent cosine potential, combined with the compact topology of the dual sector, yields profound implications for high-density matter, unifying gravitational and nuclear phenomena within the same torsional framework and providing a self-consistent description of compact objects without singularities or event horizons.

6.1 Nuclear Forces as the First Torsional Layer Cycle

At nuclear saturation density ($\rho \approx \rho_c$), the coupling $\mu(\rho_c) \approx \mu_0$ becomes sufficient to lock the relative phase $\Delta\phi \approx \pi$, marking the first minimum of the cosine potential beyond the vacuum anti-synchronization state. This shift dominates the dual-sector rotation contribution in Ω_{biv} , flipping the sign of $J < 0$ and producing short-range repulsion—the observed hard-core in nucleon-nucleon scattering at ~ 0.5 fm.

As density increases slightly inward, access to the next cosine minimum ($\Delta\phi \approx 3\pi$ or equivalent winding) restores $J > 0$ with amplified magnitude, yielding strong attraction responsible for nuclear binding (~ 40 – 50 MeV per nucleon). This single mechanism reproduces both the repulsive core and

attractive well of the nuclear force as the *first cycle* of the infinite torsional layering: repulsion (outer shell, $J < 0$) followed by deeper attraction ($J > 0$).

Pauli exclusion emerges dynamically from phase rigidity in the repulsive layer, while charge independence and spin-orbit effects arise from resonance conditions in the dual rotor spectrum. The strong nuclear force is thus demoted to gravitational torsion in its layered, high-density phase—no fundamental SU(3) gauge group is required.

6.2 Atomic Nuclei as Primordial Torsion Stars

Every atomic nucleus is a microscopic torsion star: a compact, chaotic condensate exhibiting the same alternating layer structure on femtometer scales. The central region features ultra-strong attractive zones binding quarks and nucleons, surrounded by repulsive shells enforcing confinement and surface tension. The observed saturation of nuclear density, magic numbers, and limits on the nuclear chart ($A \gtrsim 300$) follow from torsional resonance stability and spontaneous bounce in over-wound configurations.

Nuclear reactions—fission as explosive layer flip to repulsion, fusion as tunneling through the outer barrier—emerge naturally from cosine potential transitions.

6.3 Layered Structure of Torsion Stars

In supra-nuclear regimes ($\rho \gg \rho_c$), successive windings ($n \gg 1$) access ever-deeper cosine minima, generating infinite alternating layers of attraction ($J > 0$) and repulsion ($J < 0$) with increasing amplitude toward the center. This self-regulating structure forbids central singularities: collapse is halted and reversed repeatedly, settling into a quasi-stable turbulent condensate.

The central region approaches a “floating core” where isotropic phase cancellation yields $J \approx 0$, mimicking torsion-free vacuum despite extreme density. No event horizon forms; instead, a quasi-horizon arises from infinitely steep integrated torsional barriers, allowing modified Hawking-like radiation via rotor tunneling while preserving information in dual phases.

6.4 Reinterpretation of Neutron Stars and Pulsars

Observed neutron stars are torsion stars with multiple active layers. Pulsar radio pulses and timing properties originate not from rotation-powered magnetism but from high-frequency longitudinal oscillations of the dual phase $\Delta\phi$ (Josephson plasma modes) propagating through layered shock zones, modulated by interior torsional turbulence.

Glitches, period derivatives, magnetar flares, and fast radio bursts reflect violent layer transitions or winding adjustments releasing torsional strain. The maximum mass ($\sim 2.5M_\odot$) is capped by torsional layering stability, explaining the absence of heavier pulsars without exotic matter. Gravitational-wave signals from mergers feature characteristic echoes from internal layer reflections.

These consequences resolve longstanding paradoxes—nuclear force unification, singularity avoidance, pulsar energetics—while predicting measurable deviations from standard GR/NS models, testable with future NICER, LIGO, and multi-messenger observations.

The quantitative predictions and numerical verification of these structures are discussed in the next section.

7 Numerical Implications and Observational Predictions

The cosine potential hypothesis, with its density-dependent coupling $\mu(\rho)$, lends itself to quantitative testing through numerical solution of the static, spherically symmetric field equations derived from the torsional mismatch scalar J . These computations will constrain the free parameters (μ_0, γ) and yield falsifiable predictions for nuclear physics, neutron-star properties, and multi-messenger astrophysics.

7.1 Spherically Symmetric Static Solutions

In spherical symmetry, the rotor angles $\theta(r)$ and $\phi(r)$ satisfy coupled ordinary differential equations obtained from minimizing the action density (or equivalently from the teleparallel equations with the cosine-driven phase selection). The system reduces to

$$\frac{d\theta}{dr} = f(\theta, \phi, \rho(r); m, \mu(\rho)),$$

$$\frac{d\phi}{dr} = g(\theta, \phi, \rho(r); \mu(\rho)),$$

supplemented by the density profile $\rho(r)$ sourced self-consistently by the local torsional energy (analogous to the Tolman–Oppenheimer–Volkoff equations but with layered torsion).

Ongoing numerical work (publicly available at <https://github.com/hypernumbernet/dual-spacetime-simulator>) integrates these equations outward from a floating core ($J \approx 0$, $\theta \approx \phi \pmod{2\pi}$) using adaptive stepping to resolve sharp layer transitions. Initial phenomenological runs with $\gamma = 4$ and μ_0 tuned to nuclear binding already reproduce:

- Nuclear saturation at ρ_c with surface repulsion,
- Binding energy per nucleon ~ 40 – 50 MeV in the first attractive layer,
- Sharp density drop at nuclear radii.

Full N-body or mean-field integrations incorporating baryonic distributions are in preparation.

7.2 Equation of State and Neutron-Star Mass-Radius Relation

The layered torsional structure yields a stiff equation of state (EOS) above ρ_c , with oscillatory pressure contributions from successive repulsive layers. This predicts:

- Sound speed exceeding $c/\sqrt{3}$ in deep layers (without causality violation, as torsion regulates propagation),
- Maximum mass ~ 2.5 – $2.8M_\odot$ (capped by winding instability, explaining the absence of stable objects beyond $\sim 3M_\odot$),
- Radius ~ 12 – 14 km for $2M_\odot$ stars, consistent with NICER constraints but with distinctive multi-layer imprints in tidal deformability.

Deviations from standard hadronic/quark EOS appear as pressure peaks at integer multiples of ρ_c , detectable via precise mass-radius measurements (future NICER upgrades, Athena).

7.3 Gravitational-Wave Echoes and Merger Signatures

Binary torsion-star mergers produce gravitational waves with post-merger echoes from internal layer reflections, at frequencies tied to the plasma-like torsional oscillation spectrum ($\omega \propto \sqrt{\mu(\rho)}$). These quasi-periodic modulations (period ~ 1 – 10 ms) distinguish torsion stars from black holes, already hinted in some LIGO/Virgo reanalyses.

The absence of clean ringdown (replaced by chaotic torsional turbulence) and potential repetitive bursts from layer excitations offer smoking-gun signatures for next-generation detectors (Einstein Telescope, Cosmic Explorer).

7.4 Pulsar Timing and Emission

Pulsar pulses arise from beamed longitudinal torsional waves rather than rotation. This predicts:

- Minimum periods limited by Planck-scale dual frequency (far below observed ms),
- Glitches as sudden winding adjustments,
- Negative period derivatives in some systems (frequency increase via core contraction),
- Coherent radio emission via curvature radiation in torsional shock zones.

Fast radio bursts and magnetar flares correspond to violent layer flips, releasing stored torsional energy without extreme magnetic fields.

These predictions are immediately testable with SKA, FAST, and continued pulsar monitoring, providing near-term discrimination from conventional neutron-star models.

The concluding section summarizes the framework and outlines future directions.

8 Conclusions and Outlook

In this paper, we have presented a careful extension of the double spacetime theory to the strong-field regime through the introduction of a density-dependent cosine potential governing the relative phase between usual and dual rotors. This hypothesis resolves the key open question of how the dual rotation angle ϕ comes to dominate the usual rapidity θ at high densities, naturally generating the infinite alternating attractive-repulsive layers predicted by the original framework.

Crucially, the proposal preserves the algebraic elegance and background independence of the parent theory: the torsional mismatch scalar J , the rotor product $R_{\text{total}} = R_{\text{usual}}R_{\text{dual}}$, and the action remain unchanged, ensuring exact equivalence with teleparallel gravity and General Relativity in all regimes where the cosine term is negligible. The potential merely provides a physically motivated microscopic mechanism—collective resonance among particle-intrinsic dual spacetimes, deeply analogous to Josephson coupling in superconductivity—that selects the density-dependent values of $\theta(r)$ and $\phi(r)$.

The resulting picture is profoundly unifying: nuclear forces emerge as the first torsional layer cycle, atomic nuclei as primordial torsion stars, and neutron stars as multi-layered torsional superconductors forbidding singularities and event horizons. Dark matter is rendered unnecessary, inertia and gravity remain identical phenomena, and the continuum hypothesis is comprehensively abandoned.

Looking ahead, several promising directions present themselves. Full numerical integration of the spherically symmetric equations, already underway in open-source simulators, will precisely constrain μ_0 and γ against nuclear data and neutron-star observations, yielding detailed mass-radius relations and gravitational-wave templates with distinctive echo signatures.

Experimental verification may prove feasible in the near term through superconducting metamaterials designed to mimic high-density dual resonance, potentially demonstrating modest torsional modulation at laboratory scales—a first step toward gravitational engineering via coherent excitation of dual phases.

Finally, the compact $SU(2)$ structure of the dual sector invites natural extension to quantum gravity: dual rotor phases directly parameterize Hilbert space states, with torsional mismatch encoding interactions and de Broglie waves emerging from intrinsic oscillator dynamics. This promises a fully algebraic unification of all forces within the 16-dimensional biquaternion algebra, without extra dimensions or ad-hoc quantization.

The double spacetime framework, augmented by this cosine potential, offers not merely a modification of General Relativity but its completion: a theory in which gravity is demoted to an engineerable torsional degree of freedom, singularities are forbidden by self-regulation, and the microscopic origin of spacetime itself is revealed as paired, particle-local, and discretely layered.

We anticipate that ongoing numerical and experimental work will soon provide decisive tests, potentially ushering in a new era of gravitational understanding and control.